NATIONAL ECONOMICS UNIVERSITY
COLLEGE OF TECHNOLOGY

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MID-TERM ASSIGNMENT

COURSE: Data Analysis with Python

TOPIC: CLUSTERING EMPLOYEES BY WORK BEHAVIOUR AND SATISFACTION – HR ANALYTICS CASE STUDY

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Abstract

This report applies unsupervised learning to the *HR Analytics Case Study* dataset, which describes about 4,410 employees of a company through demographic, job, survey and attendance (swipe-in/swipe-out) records.

The goal is to discover naturally occurring employee segments without using any pre-defined label. We integrate the relational files, clean and engineer features (including average daily working hours derived from attendance logs), and compare three clustering algorithms — K-Means, Hierarchical (Ward) clustering and DBSCAN — using the Silhouette score, the Elbow method, Davies–Bouldin and Calinski–Harabasz indices.

The best model is [best model, k = ?] with a Silhouette score of [value]. The resulting segments are profiled and cross-checked against the `Attrition` column (used only for external validation) to derive practical HR recommendations.

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1 Initial Data Inspection

Before data preprocessing, an initial inspection was conducted to understand the structure and quality of all five datasets. The inspection focused on the consistency of the files, the relationships between them, file encoding, data types, missing values, and duplicate records.

1.1 Dataset Overview

The HR Analytics dataset consists of five relational files: `general`, `employee`, `manager`, `in_time`, and `out_time`. Each file contains 4,410 records.

Table 1 summarizes the basic information of the five datasets.

Table 1: Overview of the five datasets

Criteria	General	Employee	Manager	In-time	Out-time
Records	4,410	4,410	4,410	4,410	4,410
Attributes	24	4	3	262	262
Missing values	28	83	0	109080	109080
Duplicate records	0	0	0	0	0

Across the five files, there are 555 attributes in total. However, the 22,050 records represent the sum of records across files, not the number of unique employees, since the same employees are represented in multiple relational files.

1.2 Column Consistency

The five files describe different information, so they cannot be concatenated; they must be merged later. Consistency was therefore checked in two ways. First, the numbers of rows and columns were compared (Table 1): every file has 4,410 rows, while the number of columns differs by file (24, 4, 3, 262 and 262). Second, the column names of `in_time` and `out_time`, which have the same structure (one column per calendar day), were compared. They contain [the same 262 column names in the same order], so the two files can be processed in parallel when computing working hours. No column name other than `EmployeeID` is shared between the other files [confirm], so no naming conflict (e.g. `_x`, `_y` suffixes) will appear after merging.

1.3 Relationships Between Files

The common column of the five files is `EmployeeID`. In `in_time` and `out_time` this column is stored in an unnamed first column (`Unnamed: 0`) and is renamed to `EmployeeID`. Table ?? shows the key checks.

Table 2: Primary key / foreign key check on `EmployeeID`

File	Null keys	Duplicate keys	IDs not in General	General IDs missing
General	[0]	[0]	[0]	[0]
Employee	[0]	[0]	[0]	[0]
Manager	[0]	[0]	[0]	[0]
In-time	[0]	[0]	[0]	[0]
Out-time	[0]	[0]	[0]	[0]

`EmployeeID` is the **primary key** of the `general` dataset (unique and non-null) and the **foreign key** in the other four files, so the relationship is one-to-one. A test merge of `general`, `employee` and `manager` on `EmployeeID` returned [4,410] rows, i.e. no record was lost or duplicated.

1.4 File Encoding

The encoding of each CSV file was tested by decoding the raw bytes. [All five files were successfully decoded as UTF-8 (no BOM, no decoding error)], so no special encoding argument is needed when calling `pd.read_csv()` and no font or character errors are expected.

1.5 Data Types

The data types of each dataset were examined using the `info()` method.

Table 3: Distribution of data types across datasets

Dataset	int64	float64	str	Total attributes
General	14	2	8	24
Employee	1	3	0	4
Manager	3	0	0	3
In-time	1	12	249	262
Out-time	1	12	249	262
Total	**20**	**29**	**506**	**555**

The `general` dataset contains demographic and job-related information. The `employee` dataset provides employee satisfaction and work-life balance information, while the `manager` dataset contains job involvement and performance ratings.

The `in_time` and `out_time` datasets each contain 262 attributes, including 249 string attributes, 12 floating-point attributes, and 1 integer attribute. These two files contain attendance-related records and will be examined further during feature engineering.

1.6 Missing Values

Missing values were identified in the `general` and `employee` datasets.

1.6.1 General Dataset

The `general` dataset contains 28 missing values across two attributes.

Table 4: Missing values in the General dataset

Attribute	Missing values	Percentage
NumCompaniesWorked	19	0.43%
TotalWorkingYears	9	0.20%
Total	**28**	**0.63%**

1.6.2 Employee Dataset

The `employee` dataset contains 83 missing values across three attributes.

Table 5: Missing values in the Employee dataset

Attribute	Missing values	Percentage
EnvironmentSatisfaction	25	0.57%
JobSatisfaction	20	0.45%
WorkLifeBalance	38	0.86%
Total	**83**	**1.88%**

1.6.3 Manager Dataset

The `manager` dataset contains three attributes: `EmployeeID`, `JobInvolvement`, and `PerformanceRating`. All 4,410 records are non-null, indicating that no missing values were detected.

1.6.4 Attendance Datasets

The `in_time` and `out_time` datasets each contain 262 attributes and 4,410 records. The available inspection output does not provide the non-null counts for individual columns. Therefore, their exact numbers of missing values need to be calculated separately before preprocessing.

1.7 Duplicate Records

Duplicate records were checked in all five datasets. The results showed that no duplicate records were detected in any of the files.

Table 6: Duplicate record check

Dataset	Number of duplicate records
General	0
Employee	0
Manager	0
In-time	0
Out-time	0

1.8 Initial Data Quality Assessment

Based on the initial inspection, the following observations were identified:

- All five datasets contain 4,410 records each, with a total of 555 attributes.
- The datasets contain numerical and string data types, requiring appropriate preprocessing techniques.
- The `general` dataset contains 28 missing values, while the `employee` dataset contains 83 missing values.
- The `manager` dataset contains no missing values.
- The exact number of missing values in the attendance datasets still needs to be determined.
- No duplicate records were detected in the five datasets.
- `EmployeeID` links all files (primary key in `general`, foreign key elsewhere), and the files can be read with UTF-8 encoding.

These findings provide an overview of the data structure and initial data quality. The next step is to examine the attendance records, handle missing values, and prepare suitable features for clustering.

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